

Q- Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.

MPLS (**Multiprotocol Label Switching**) is a networking technology that forwards data using short **labels** instead of checking the full IP routing information at every router.

In traditional IP routing, each router examines the destination IP address and independently chooses the next hop using its routing table.

With MPLS, packets are assigned to a **Forwarding Equivalence Class (FEC)** and sent through a predefined **Label Switched Path (LSP)**.

Routers can then forward packets by looking at the MPLS label, making forwarding more predictable and efficient.

MPLS also supports **traffic engineering**, allowing network operators to control paths and manage network resources more effectively.

Therefore, MPLS is commonly used in enterprise WANs where reliable, predictable connectivity between different locations is important.